

## Integration — Lesson 33 Test: Improper Integrals

10 questions · show your work in the box · write "converges to \_\_\_\_" or "diverges" with the limit that proves it

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_ / 10

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1. Find each limit ("where is it heading?"): (a)  $\lim_{b \rightarrow \infty} 1/b^2$  (b)  $\lim_{b \rightarrow \infty} (4 + 1/b)$  (c)  $\lim_{b \rightarrow \infty} \ln b$  1 pt

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2. Evaluate  $\int_1^{\infty} 1/x^5 \, dx$  using the three-step method (wall at b, integrate, let  $b \rightarrow \infty$ ). 1 pt

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3. Evaluate  $\int_3^{\infty} 1/x^2 \, dx$ . (Careful — the lower limit is 3, not 1.) 1 pt

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4. Show that  $\int_1^\infty 1/x^{1/3} dx$  diverges, and explain in one sentence why it diverges even though  $1/x^{1/3} \rightarrow 0$ .

1 pt

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5. Evaluate  $\int_0^\infty e^{-3x} dx$ .

1 pt

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6. Use the p-rule to decide convergent or divergent — and give the value if convergent: (a)  $\int_1^\infty 1/x^6 dx$  (b)  $\int_1^\infty 1/x^{0.8} dx$

1 pt

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7. The function  $1/\sqrt{x}$  blows up at  $x = 0$ . Evaluate  $\int_0^9 1/\sqrt{x} dx$  using a wall at  $x = a$  with  $a \rightarrow 0^+$ .

1 pt

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8. Show that  $\int_0^1 1/x \, dx$  diverges by computing the limit as the lower wall  $a \rightarrow 0^+$ .

1 pt

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9. In your own words (2–3 sentences): what does it mean for an improper integral to CONVERGE, and how can a region that is infinitely long have a finite area? You may use  $\int_1^\infty 1/x^2 \, dx = 1$  as your example.

1 pt

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10. Challenge: evaluate  $\int_2^\infty 1/(x(x-1)) \, dx$ . (Hint: partial fractions first, then combine the logs into one before taking the limit.)

1 pt

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